

Title

Model inputs, outputs, and scripts associated with “Continental-scale controls on hyporheic respiration revealed by knowledge-guided machine learning”

Summary

This data package is associated with the publication “Continental-scale controls on hyporheic respiration revealed by knowledge-guided machine learning” published in Environmental Research Communications (Zheng et al. 2026; DOI: [10.1088/2515-7620/ae9525](https://doi.org/10.1088/2515-7620/ae9525)). The project combines mechanistic process modeling with knowledge-guided machine learning (KGML) to evaluate how organic matter chemistry, microbial biomass, and physical substrate accessibility regulate realized respiration rates across river corridors.

All data used in this paper have been previously published and can be accessed at <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/1729719> (Goldman et al., 2020). This data package contains the preprocessing scripts for the previously published data and subsequent modelling workflows. The full workflow with input and output data can be found in the associated GitHub repository at <https://github.com/river-corridors-sfa/KGML-WHONDRS>.

A preliminary version of this data package was published in January 2026 at the time of manuscript submission. It was updated in August 2026, at the time of manuscript acceptance, to include additional metadata (this readme, data dictionary, and file level metadata) and include the final scripts that were revised during manuscript revisions.

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

Brief Overview of Methods

Hyporheic sediment samples were collected as part of the WHONDRS S19S field campaign across multiple river corridors in North America. Laboratory incubations were conducted to quantify sediment respiration rates under controlled conditions. Microbial biomass, organic matter chemistry, and physical sediment properties were measured using standard biogeochemical and analytical protocols, and site-level metadata were compiled from field observations (details can be accessed via ESS-DIVE: [doi:10.15485/1729719](https://doi.org/10.15485/1729719)).

The dataset supports mechanistic and machine-learning modeling workflows. A process-based respiration model was implemented using thermodynamically informed rate formulations with explicit representation of organic matter chemistry, microbial biomass, and physical substrate accessibility. Knowledge-guided machine learning (KGML) workflows incorporated mechanistic model outputs to structure model inputs and residual learning, supporting reproducible integration of process-based and data-driven methods.

Critical Details

1 – The raw data is not included in the data package on ESS-DIVE because it has been previously published. The full workflow with input and output data can be found in the associated GitHub repository at <https://github.com/jianqiuz/KGML-WHONDRS>.

2 – Reproducibility Notes:

- All derived outputs are written to derived/.
- Reviewer-facing figures are written to figures/.
- Data-processing QC figures use the process_*.pdf prefix.
- Cross-validation is seeded.
- Leakage-sensitive operations are kept inside fold loops where possible.
- Site-grouped cross-validation is used by default for KGML so upstream/midstream/downstream samples from the same site do not appear in both training and testing folds.

Data Package Structure

In addition to this readme, this data package also includes a file-level metadata (FLMD) file that describes each file and a data dictionary (DD) that describes all column/row headers and variable definitions.

This data package consists of (1) readme; (2) data dictionary (dd); (3) file level metadata (flmd); (4) model scripts; and (5) three folders: (1) R-scripts; (2) figures; and (3) derived data (i.e. outputs from the scripts). All files are .pdf, .csv, .Rmd, .R, .py, .png.

Citations, Acknowledgements, and License

Acknowledgements

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Cite this data package with the appropriate DOI. Cite the associated manuscript in any work that uses analyses or conclusions presented in the manuscript

Citations

- Goldman A E ; Arnon S ; Bar-Zeev E ; Chu R K ; Danczak R E ; Daly R A ; Delgado D ; Fansler S ; Forbes B ; Garayburu-Caruso V A ; Graham E B ; Laan M ; McCall M L ; McKeever S ; Patel K F ; Ren H ; Renteria L ; Resch C T ; Rod K A ; Tfaily M ; Tolic N ; Torgeson J M ; Toyoda J G ; Wells J ; Wrighton K C ; Stegen J C ; WHONDRS Consortium T (2020): WHONDRS Summer 2019 Sampling Campaign: Global River Corridor Sediment FTICR-MS, Dissolved Organic Carbon, Aerobic Respiration, Elemental Composition, Grain Size, Total Nitrogen and Organic Carbon Content, Bacterial Abundance, and Stable Isotopes v8). River Corridor and Watershed Biogeochemistry SFA, ESS-DIVE repository. Dataset. doi:10.15485/1729719
- Zheng, J., Garayburu-Caruso, V., Scheibe, T., & Stegen, J. (2026). Continental-scale controls on hyporheic respiration revealed by knowledge-guided machine learning. Environmental Research Communications, 8(8), 081011. [doi:10.1088/2515-7620/ae9525](https://doi.org/10.1088/2515-7620/ae9525)

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Change History

Data Package Version	Changes
Preliminary <i>January 2026</i>	Preliminary data package publication
Version 1 <i>October 2026</i>	<u>NEW FILES</u> All files are new